

Ambiguity, Investor Composition, and Sovereign Rollover Crises*

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Abstract

Rollover crises depend on more than fundamentals: they depend on who holds the debt and on how far those holders trust official information about it. I put creditors with multiple priors about the bias of official information into a global game of sovereign debt rollover and obtain three exact results. Distrust is priced like bad news: a creditor with ambiguity radius δ behaves as a Bayesian who heard official news worse by δ , and the premium in his run cutoff equals δ times the ratio of public to private precision, so distrust bites hardest exactly where investors must lean on official data. Equilibrium uniqueness is unchanged: ambiguity moves the crisis threshold while preserving the selected global-game equilibrium, which makes fragility a smooth, measurable function of the share of debt held by distrustful, price-sensitive investors, a share that investor-base data track directly. Official communication splits into two instruments that current policy conflates: precision, which stabilizes only when the news is good enough and destabilizes in distress, and credibility, which stabilizes always. Distrust differs from volatility: added noise drags the crisis threshold toward an uninformative-signal anchor and can stabilize in deep distress, while distrust destabilizes everywhere, so disagreement and distrust proxies enter empirical work as separate regressors with different predicted signs. In a calibration anchored to sovereign haircut and spread magnitudes, the ambiguity premium is 18 basis points in calm markets and 98 in stress. An investor-base shift of the size quantitative tightening produces prices at 10 basis points in calm and 44 on the arrival of bad news: the cost of the composition arrives exactly when it hurts.

JEL codes: D81, D82, F34, G01, H63.

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1 Introduction

In October 2009 the incoming Greek government announced that the year’s budget deficit would be about 12.7 percent of GDP: roughly double the figure reported weeks earlier, and later revised

*Revised draft: July 2026. The empirical sections specify the investor-base panels, real-time fiscal revisions, and auction data used for estimation. Every number in the text is either documented in a cited source or an output of the model solved here. Replication code in Julia and R accompanies the draft.

to 15.4 percent. The European Commission’s January 2010 report on Greek statistics documented severe irregularities in the country’s fiscal reporting (European Commission, 2010). Spreads on Greek ten-year bonds, under two hundred basis points in the summer of 2009, exceeded six hundred by April 2010. In May the country lost market access and entered an official program. The striking feature of this sequence is what moved first. Greek fundamentals deteriorated smoothly. Credibility moved discontinuously: every number the government had ever published became, overnight, a number a creditor could reasonably doubt.

This paper is about that object: the doubt itself, who holds it, and what it does to the arithmetic of debt rollover. I study a global game of sovereign rollover in the tradition of Morris and Shin (2004) and Cole and Kehoe (2000), with one addition and one state variable. The addition is ambiguity: creditors face an unknown bias in official information and evaluate rollover under the worst prior in a set, in the manner of Gilboa and Schmeidler (1989). The state variable is the investor base: the share of maturing debt held by creditors who carry that distrust, as against holders, domestic banks and official lenders above all, who carry a zero distrust margin. Both objects are measurable. The distrust has natural proxies in the history of fiscal data revisions and in transparency standards. The composition is tracked country by country in the data of Arslanalp and Tsuda (2014). The paper’s claim is that their interaction prices rollover risk.

Three exact results organize the theory. The first is an as-if theorem with a useful loading. A creditor whose prior over the official signal’s bias spans $[-\delta, \delta]$ and who evaluates the rollover decision at the worst case behaves exactly as a Bayesian who received official news worse by δ : ambiguity is priced as pessimism, as in Ilut and Schneider (2014), but here inside a coordination game where the pessimism is contagious through the run margin. The loading is the sharp part: the premium in the creditor’s run cutoff is $(\alpha/\beta)\delta$, the ambiguity radius times the ratio of public to private signal precision. Distrust of official data is priced in proportion to how much the creditor must rely on official data. Sovereigns whose fiscal position is illuminated by little private information, few analysts and thin markets, pay the most for a given loss of credibility, and improving private information about a country is a substitute for trusting its government.

The second result is uniqueness with a moving threshold. The global-game uniqueness condition, that public precision be bounded relative to the square root of private precision (Hellwig, 2002; Morris and Shin, 2003), is unchanged by any profile of ambiguity radii across creditors, because distrust enters as a signal shift rather than a slope. The crisis threshold remains unique and moves smoothly, which makes fragility a continuous function of measurable composition. The comparative statics come with the fixed-point amplification familiar from coordination models, a multiplier of 2.1 at the calibration, and they interact: the effect of shifting debt toward distrustful holders is small in calm and large in stress, because the marginal creditor’s density sits where the action is. At the calibration, the ambiguity premium in spreads is 18 basis points in calm markets and 98 basis points under stress: a ratio of 5.4. Distrust is an option that pays off, against the sovereign, exactly when fundamentals disappoint.

The third result reorganizes the economics of transparency. A public signal in this model has

two attributes that policy language runs together: how precise it is, and how much its bias is trusted. Raising precision has the Morris and Shin (2002) property of empowering whatever the news says, stabilizing when the news is good relative to the crisis threshold and destabilizing when it is bad, and the distrust loading shifts the boundary: with fragile holders, precision must clear a higher bar of good news before it helps. Raising credibility, shrinking δ , stabilizes unconditionally, at every news level and every composition. The two instruments play different roles, and the model gives the exact frontier (Figure 2): official information is most dangerous, and credibility most valuable, exactly in the distressed region where governments reach for communication. A corollary sharpens the measurement. Adding variance to the public signal drags the crisis threshold toward its uninformative-signal anchor $1 - \bar{p}$, and can therefore *stabilize* a sovereign in deep distress by diluting bad news, while distrust destabilizes at every news level. Volatility, disagreement, and distrust therefore enter empirical work separately, with different predicted signs in distress: a discipline the designs of Section 7 implement.

The policy experiments give the theory its current-events content. Quantitative tightening, in this model, is a composition treatment: the central bank, a holder with zero run margin, hands maturing debt back to price-sensitive investors with positive δ . The experiment prices the handoff: drifting the fragile share from 0.35 to 0.65 costs 10 basis points while fundamentals are calm, and 44 basis points the day moderate bad news arrives. The composition is an option position: cheap to carry, expensive exactly in the state it was supposed to insure. Collective action clauses, which raise expected recovery in restructuring, work here as coordination policy: raising recovery from 0.60 to 0.70 cuts the stress spread from 575 to 128 basis points and, notably, shrinks the ambiguity premium itself, because a smaller loss given default makes the worst case less frightening. And official backstops act as ambiguity floors: a facility that truncates the worst case does disproportionate work when δ is high, which is a multiple-priors reading of when catalytic official finance works (Morris and Shin, 2006).

The paper builds on four literatures and differs from each in one identified way. The global-games tradition (Morris and Shin, 2003; Morris and Shin, 2004; Hellwig, 2002) supplies the equilibrium machinery. I add heterogeneous multiple-priors creditors and show the selection survives. The ambiguity literature supplies the preferences: Gilboa and Schmeidler (1989) the decision rule, Epstein and Schneider (2008) ambiguous information quality in asset pricing, Ilut and Schneider (2014) as-if pessimism in macro, Caballero and Krishnamurthy (2008) Knightian flight in crises. This paper puts these creditors into a coordination game with a composition state variable. Ui (2025), the closest theoretical neighbor, studies strategic ambiguity in symmetric global games. The sovereign-crisis literature (Calvo, 1988; Cole and Kehoe, 2000; Bocola and Dovis, 2019) treats rollover risk with Bayesian creditors, where fragility lives in multiplicity. Here fragility lives in a unique, moving threshold, which is a different and more testable object. And the empirical investor-base literature, anchored by Arslanalp and Tsuda (2014), documents the composition shifts the theory prices but has lacked a mechanism connecting them to crisis thresholds. The bridge between those data and this game is, I would argue, the paper’s most useful contribution.

The empirical strategy turns the model’s primitives into measured objects. Distrust is recovered from revision histories, transparency standards, dispersion measures, and the structural inversion of spreads into a country-level $\delta_{i,t}$. Composition comes from investor-base panels. The dynamic block links static games through slow composition and credibility states, which is enough to price quantitative tightening, CACs, and backstops. Domestic banks enter the baseline as stable holders, and the sovereign-bank nexus enters the extension as the balance-sheet feedback that converts spread movements into fundamentals.

My plan is as follows. Section 2 sets out the game and the ambiguity results. Section 3 adds the investor base. Section 4 treats official information. Section 5 runs the policy experiments. Section 6 reports the calibrated magnitudes. Section 7 specifies the empirical designs. Section 8 concludes. Proofs and extensions are in the appendix.

2 The Rollover Game

The analytical structure is exact inside the Gaussian one-shot game. Propositions 1 through 5 follow from normal signals, max-min preferences over a bias interval, Assumption 1, and the uniqueness inequality. Section 6 attaches magnitudes to those objects with documented payoffs, investor-base shares, information precisions, and revision-based distrust measures. Section 5 chains the static game into the policy experiments.

2.1 Environment

A sovereign has a unit continuum of short-term creditors, each holding one maturing bond. Each creditor chooses to roll over or to run. The sovereign survives the rollover iff the mass of runners is at most its liquid capacity θ : fundamentals are the fraction of maturing debt the sovereign could redeem using buffers and primary surplus. If $\theta < 0$ the sovereign enters crisis regardless, and if $\theta > 1$ it survives even a full run. Between, survival is a coordination outcome.

Payoffs are in units of face value. Rolling over returns the promised gross yield $R > 1$ if the sovereign survives and the recovery $\mathcal{R} < 1$ in crisis. Running means selling into the secondary market at the fire-sale price $1 - \kappa$. A creditor therefore rolls over when his assessed crisis probability P satisfies

$$(1 - P)R + P\mathcal{R} \geq 1 - \kappa \quad \Longleftrightarrow \quad P \leq \bar{p} \equiv \frac{R - (1 - \kappa)}{R - \mathcal{R}}, \quad (1)$$

the hurdle probability that summarizes the security’s cash flows: at the calibration, $\bar{p} = 27.3$ percent.

Information has a public and a private layer. All creditors see the official signal $y = \theta + \sigma_y \eta$, precision $\alpha = \sigma_y^{-2}$: the budget documents and the program reviews. Each creditor i also has private information $x_i = \theta + \sigma_x \varepsilon_i$, precision $\beta = \sigma_x^{-2}$, with ε_i independent standard normal: his own analysis and client flows.

2.2 Distrust as multiple priors

The departure is that creditors discount the official signal. Type- k creditors entertain the set of models in which the official signal carries an unknown bias b ,

$$y = \theta + b + \sigma_y \eta, \quad b \in [-\delta_k, \delta_k],$$

and evaluate actions by the worst model in the set (Gilboa and Schmeidler, 1989): the max-min creditor asks what the rollover is worth if the books are cooked against him by as much as his doubt allows. The radius δ_k is the paper’s measure of distrust: zero for a creditor who accepts official statistics, large after an episode like Greece’s 2009 revisions. It is a primitive about the credibility of the *information source*, rather than about risk appetite, and Section 7 ties it to observable revision histories.

The worst case is one-directional because the exit is safe, and since the as-if theorem leans on this, I state it as an assumption rather than leave it implicit.

Assumption 1 (Model-free exit). *The run payoff $1 - \kappa$ is independent of the unknown bias b : a creditor who exits collects a posted secondary-market price insulated from official statistics.*

The assumption is substantive. If the fire-sale price itself depended on fundamentals, or worsened in the crisis event, the runner would face his own worst case, a second minimization would appear, and the clean subtraction of Lemma 1 would pick up a correction term. The one-shot secondary-market sale at a posted discount is the case where the assumption holds exactly. Where exit prices are themselves state contingent, the results below survive as long as the rollover payoff is more exposed to the worst case than the exit payoff, which is what short maturity buys the runner. With the assumption in place, the game resolves the worst case cleanly.

Lemma 1 (Worst-case selection and as-if pessimism). *In any equilibrium in threshold strategies, the crisis event is $\{\theta \leq \theta^*\}$ for some θ^* . The worst-case model for a rolling creditor of type k is $b = \delta_k$: official news overstates fundamentals by the full radius. Type k therefore behaves exactly as a Bayesian creditor who observed the public signal $y - \delta_k$.*

The lemma does two jobs. It disposes of a circularity worry, because the crisis event in a threshold equilibrium is a fundamental event, and the worst case is selected once before the fixed point is computed. And it converts ambiguity into geometry: each type reads the same announcement and subtracts its own doubt, so a single official statement becomes a family of effective signals $\{y - \delta_k\}$, fanned out by distrust. I return to that fan in Remark 1, because it is observable.

2.3 The ambiguity premium and the survival of uniqueness

Bayesian updating from the effective signal gives type k ’s posterior $\theta | x_i \sim N(m_k(x_i), (\alpha + \beta)^{-1})$ with $m_k(x) = [\alpha(y - \delta_k) + \beta x]/(\alpha + \beta)$. Threshold strategies are cutoffs x_k^* : run below, roll

above. The indifference condition at the cutoff, $\mathbb{P}(\theta \leq \theta^* \mid x_k^*) = \bar{p}$, solves to

$$x_k^* = x_B^*(\theta^*) + \frac{\alpha}{\beta} \delta_k, \quad (2)$$

where x_B^* is the common Bayesian cutoff. This is the paper's first exact result.

Proposition 1 (The ambiguity premium and its loading). *Type k 's run cutoff exceeds the Bayesian cutoff by $(\alpha/\beta) \delta_k$: distrust makes creditors run at better private information, in proportion to the ratio of public to private precision. Equivalently, in signal-quantile units, type k 's run propensity at any fundamental is shifted by $(\alpha/\sqrt{\beta}) \delta_k$.*

The loading α/β carries the economics. Distrust of the official source matters only insofar as the posterior leans on that source: where private information is abundant (β large), creditors cross-check the government and δ is nearly free. Where the official account is all there is, every unit of doubt translates into cutoff. This is a cross-country prediction, stated as P2 in Section 7, and it inverts a comfortable intuition: the sovereigns most exposed to a credibility loss are precisely those whose information environment official sources dominate.

Aggregating, the mass of runners at fundamental θ is $\sum_k \mu_k \Phi(\sqrt{\beta}(x_k^* - \theta))$ for composition weights μ_k , and the crisis threshold solves the fixed point

$$\theta^* = \sum_k \mu_k \Phi\left(g(\theta^*) + \frac{\alpha}{\sqrt{\beta}} \delta_k\right), \quad g(\theta^*) \equiv \frac{\alpha(\theta^* - y) - \Phi^{-1}(\bar{p})\sqrt{\alpha + \beta}}{\sqrt{\beta}}. \quad (3)$$

Proposition 2 (Uniqueness is ambiguity-proof). *If $\alpha/\sqrt{\beta} < \sqrt{2\pi}$, equation (3) has a unique solution, and the threshold equilibrium is the unique equilibrium in monotone strategies, for every composition (μ_k) and every profile of ambiguity radii (δ_k) . Ambiguity shifts the map while leaving its slope controlled by the same precision ratio.*

The condition is the familiar one (Hellwig, 2002; Morris and Shin, 2003), and its economic content is independence from δ . Distrust operates like a family of parallel signal translations, so heterogeneous ambiguity radii preserve the slope condition that selects the global-game equilibrium. This locates fragility in a moving unique threshold rather than in equilibrium selection: crises here are movements of θ^* into the region where fundamentals sit, which is why composition and distrust can be priced continuously in spreads. The comparative statics are therefore well defined objects to estimate. The inequality is verified at the calibration with room to spare, $\alpha/\sqrt{\beta} = 1.35$ against the bound of 2.51. Dispersion data discipline β and revision variances discipline α , so each sovereign's distance from the uniqueness frontier is estimable. As the frontier approaches, the multiplier $1/(1 - \Phi')$ grows, making the amplification statistic itself an early-warning statistic.

I report the condition across the full precision grid used in the quantitative section. On a twelve-cell grid spanning private noise from 0.10 to 0.30 and public noise from 0.25 to 0.50, the ratio $\alpha/\sqrt{\beta}$ runs from 0.40 to 4.80 and 3 cells cross the bound, all in the sharp-public, noisy-private corner. At the baseline public noise, private noise can rise to 0.157 before selection

appears at the sharpest public signal considered, and to 0.279 at the baseline one. The stress premium exceeds the calm premium in every unique cell. Its strength varies: the stress-to-calm ratio runs from 1.3 where the public signal is very noisy, so that both states carry sizeable premia, to effectively unbounded where high precision makes the calm state crisis-free. The grid holds the signal levels fixed, as the model does, so the calm anchor moves from 0 to 865 basis points across unique cells. Design 4 uses dispersion data to select the country-level point on this grid.

Remark 1 (Polarization is the fingerprint of distrust). *By Lemma 1, one announcement y enters type k 's posterior as $y - \delta_k$. Heterogeneous distrust therefore makes posterior means fan out in response to public information: the cross-sectional dispersion of expectations widens with $\alpha/(\alpha + \beta)$ times the dispersion of δ_k . Noise does the opposite, common noise moves everyone together. Surveys of investor expectations around official releases can tell these apart: announcements that raise disagreement while moving the mean are the signature of distrust rather than ordinary information. This is P6 in Section 7.*

What the static game leaves open is who the types are and how many of each hold the debt. That composition is the paper's state variable, and the next section makes it the comparative-static of interest.

3 The Investor Base

Every proposition so far holds for any finite set of types, and the sums over k in (3) are written that way. For calibration and measurement I now take two types, chosen for measurement rather than elegance, and Appendix F records how payoff heterogeneity across many classes enters through type-specific hurdles while preserving each result. *Stable* holders, share $1 - \mu$, carry a zero distrust margin worth pricing: domestic banks holding to maturity for regulatory reasons, official-sector holders with a zero run decision, long-horizon institutions whose mandates absorb mark-to-market noise. *Fragile* holders, share μ , are the price-sensitive foreign funds and dealers whose reassessments move auctions, and they carry the radius $\delta > 0$. The mapping to the categories of Arslanalp and Tsuda (2014), who decompose sovereign debt ownership into foreign banks, foreign nonbanks, foreign official, domestic banks, domestic nonbanks, and central banks, is direct enough to estimate: μ is the maturing-debt share of the categories with a live run margin.

Proposition 3 (Composition is priced, with amplification and interaction). *At the unique equilibrium of (3): (i) the crisis threshold rises with the fragile share,*

$$\frac{d\theta^*}{d\mu} = \frac{\Phi(g + \frac{\alpha}{\sqrt{\beta}}\delta) - \Phi(g)}{1 - \Phi'} > 0,$$

where $1 - \Phi'$ is one minus the slope of the aggregate map: the coordination multiplier, equal to 2.1 at the calibration. (ii) The effect of composition and the effect of distrust reinforce, $\partial^2\theta^/\partial\mu\partial\delta > 0$ wherever the fragile type's density is on the rising side of its peak. (iii) Both*

effects are state dependent, largest when fundamentals put the marginal creditor near his cutoff, so the same composition shift that is nearly free in calm markets is expensive in stress.

Part (iii) deserves its number: at the calibration, the ambiguity premium in spreads is 18 basis points when official news is comfortable and 98 when news is mediocre, a factor of 5.4, with the same δ , the same μ , the same payoffs. Investor primitives are identical across the two columns. What changed is where the marginal creditor's density sits relative to his cutoff. The empirical counterpart is a prediction about *interactions*: distrust proxies and fragile-share measures should carry little unconditional explanatory power for spreads and substantial power interacted with each other and with stress, which is P1 in Section 7, and which distinguishes the mechanism from a constant risk-premium story.

The multiplier in part (i) is the coordination analogue of objects that appear wherever fixed points amplify: cross-sectional estimates of how individual creditors respond to distrust identify the numerator, the direct effect. The equilibrium consequence is larger by $1/(1 - \Phi')$. The empirical designs below estimate the direct coefficients, and the structural step in Design 4 supplies the multiplier that converts them into equilibrium effects.

4 Official Information: Precision and Credibility

An official communication policy in this model is a pair (α, δ) : how sharp the signal is, and how far it can be doubted. The pair moves the crisis threshold through different doors, and the distinction carries all of the section's content.

Proposition 4 (Precision stabilizes conditionally, credibility unconditionally). *(i) The effect of official precision on the crisis threshold satisfies*

$$\text{sign}\left(\frac{d\theta^*}{d\alpha}\right) = \text{sign}\left(y^\dagger - y\right), \quad y^\dagger = \theta^* - \frac{\Phi^{-1}(\bar{p})}{2\sqrt{\alpha + \beta}} + \frac{\sum_k \mu_k \phi_k \delta_k}{\sum_k \mu_k \phi_k},$$

so sharper official information lowers the crisis threshold if and only if the news is good enough, $y > y^\dagger$. The hurdle $y^\dagger$ exceeds the crisis threshold itself whenever the game is run-prone ($\bar{p} < 1/2$) and rises further with the distrust-weighted composition. (ii) The effect of credibility is unambiguous: $d\theta^/d\delta_k = \mu_k \phi_k (\alpha/\sqrt{\beta})/(1 - \Phi') > 0$ for every news level, every composition, and every precision.*

Part (i) is the Morris and Shin (2002) tension in rollover clothing: precision is a megaphone, and a megaphone amplifies whatever is said, including the coordination-relevant fact that others are hearing it too. What ambiguity adds is the third term of $y^\dagger$: with distrustful holders, the megaphone must clear a higher bar of good news before it helps, because part of what it amplifies is the worst-case reading. The practical geometry is in Figure 2: the destabilizing region is the distressed corner, weak announced fundamentals and fragile composition, which is exactly the corner in which governments and official lenders most want to communicate. Official information is most dangerous where it is most demanded. Part (ii) is the clean instrument: shrinking δ ,

through statistical standards and track record, stabilizes for every news level and composition. At the calibration, halving the distrust radius in the stress calibration is worth 50 basis points of spread. The pair of results reads as an ordering of communication policy: buy credibility first, precision second, and precision only bundled with news you are confident is good. The welfare lens of Angeletos and Pavan (2007) adds distributional terms. The crisis-threshold lens here maps one-for-one to spreads.

The second result of this section separates distrust from every second-moment object it is usually proxied by.

Proposition 5 (Distrust is directional, noise is anchored). *(i) $d\theta^*/d\delta > 0$ at every news level y , every composition, and every precision: distrust always destabilizes. (ii) As official precision vanishes, the crisis threshold converges to the uninformative-signal anchor $\theta_\infty^* = 1 - \bar{p}$, so added public noise moves θ^* toward $1 - \bar{p}$: it destabilizes when news had been reassuring ($\theta^* < 1 - \bar{p}$) and stabilizes in deep distress ($\theta^* > 1 - \bar{p}$), where noise dilutes bad news. (iii) Hence in sufficiently distressed states the two comparative statics have opposite signs, and variance-based measures lose global proxy content for distrust.*

Figure 3 draws the proposition: the distrust effect is a positive, bounded ridge across all news levels, while the equal-standard-deviation noise effect crosses zero and turns stabilizing in distress. The econometric instruction is immediate: disagreement measures (forecast dispersion, bid-ask, implied-volatility smiles) and distrust measures (revision histories, transparency standards) are different regressors with different predicted coefficient *signs* in distressed subsamples. Pooling them under “uncertainty” throws away the identifying variation. P5 in Section 7 uses it.

5 Dynamics and Policy

The dynamic block is a sequence of rollover games connected by slow states: composition μ_t evolves with central-bank balance-sheet policy and fund flows, distrust δ_t with the record of official statements, fundamentals news y_t with the cycle. This block prices three experiments that map to live policy questions.

Quantitative tightening as a composition treatment. A central bank holds maturing debt with zero run margin: it is the limiting stable holder. Balance-sheet runoff hands each maturing bond back to the private margin, mechanically raising μ_t . The experiment in Section 6 drifts μ from 0.35 to 0.65 over eight years, the order of magnitude of post-2022 runoff programs relative to maturing stocks, and prices the drift twice: against calm news throughout, and against a moderate deterioration arriving in year seven. The composition premium is 10 basis points in the first case and 44 in the second. QT’s rollover cost is a state-contingent exposure, essentially all of which is collected in bad states. A debt manager who evaluates runoff at calm-market spreads is marking an option position at its premium rather than its payoff.

The doom loop. Higher spreads worsen future fundamentals through debt service, which feeds back into y_{t+1} . Appendix C shows the feedback adds a genuinely dynamic instability: past a spread level, the stable fixed point disappears and the sovereign spirals, the Calvo (1988) mechanism in modern dress, quantified for the Bayesian case by Bocola and DAVIS (2019). The headline experiments set this feedback to zero to isolate the composition and credibility mechanisms. With a positive feedback, the same wedges shrink the distance between the calm equilibrium and the spiral’s basin boundary.

Collective action clauses as coordination policy. CACs and their statutory relatives raise expected recovery $\mathcal{R}$ in restructuring. Through (1) they raise the hurdle $\bar{p}$, making creditors tolerant of more crisis risk before running, and through Proposition 5(ii) they also lower the uninformative-signal anchor $1 - \bar{p}$. Both margins are quantified in Section 6: recovery of 0.70 rather than 0.60 cuts the stress spread from 575 to 128 basis points, and shrinks the ambiguity premium itself, from 0.036 to 0.033 in threshold units, because bounding the loss bounds what the worst case can threaten. That last effect is the model’s precise version of the proposal’s conjecture that contractual design matters more in ambiguous environments: design that compresses worst-case payoffs is a substitute for trust.

Backstops as ambiguity floors. An official facility that commits to lend against fundamentals above some floor truncates the support of the worst case: a creditor may doubt the budget numbers, while the facility itself anchors the lower tail. In multiple-priors terms the backstop cuts δ directly, which by Proposition 4(ii) stabilizes unconditionally, and its value is largest exactly when δ is high, a reading of catalytic finance (Morris and Shin, 2006) in which the catalyst works on the prior set rather than the signal. The corollary for program design is uncomfortable but useful: a backstop whose own activation is discretionary re-introduces the ambiguity it was meant to remove, and prices accordingly.

6 Quantitative Magnitudes

6.1 Calibration

Table 1 lists the parameters. The payoff block is anchored to documented magnitudes: recovery of 0.60 matches mean sovereign haircuts near 40 percent (Cruces and Trebesch, 2013). The rollover yield and exit cost put the hurdle at $\bar{p} = 27.3$ percent. The information block sets private noise at 0.15 and official noise at 0.33 in fundamentals units, placing the precision ratio at 1.35, inside the uniqueness bound 2.51 with room to spare. The distrust radius $\delta = 0.08$ is disciplined by the observable that motivates the paper: official fiscal revisions of Greek magnitude, several percent of GDP against liquid capacity of order one, and the fragile share $\mu = 0.40$ sits in the range that investor-base data report for foreign nonbank holdings of emerging and peripheral sovereigns (Arslanalp and Tsuda, 2014). The news levels defining calm and stress are chosen so the calm spread is 79 basis points, an ordinary emerging-market level.

Table 1: Calibration.

Parameter		Value	Category	Basis
Rollover yield	R	1.04	fixed	promised gross return
Recovery	$\mathcal{R}$	0.60	estimated	haircuts $\approx 40\%$ (Cruces and Trebesch, 2013)
Exit cost	κ	0.08	calibrated input	fire-sale discount in stress
Hurdle	$\bar{p}$	27.3%	internal	equation (1)
Private noise	σ_x	0.15	calibrated input	fundamentals units, Design 4
Official noise	σ_y	0.33	calibrated input	fundamentals units, Design 4
Precision ratio	$\alpha/\sqrt{\beta}$	1.35	internal	uniqueness bound 2.51
Distrust radius	δ	0.08	calibrated input	revision magnitudes, Design 4
Fragile share	μ	0.40	estimated	Arslanalp and Tsuda (2014), Design 1
News, calm / stress	y	1.10 / 0.95	internal	calm spread = 79 bp

Table 2: The ambiguity premium, calm and stress.

	Calm ($y = 1.10$)		Stress ($y = 0.95$)	
	Bayesian	Distrust	Bayesian	Distrust
Crisis threshold θ^*	0.377	0.412	0.549	0.584
Crisis probability (%)	1.5	2.0	11.5	13.6
Spread (bp)	61	79	477	575
Ambiguity premium (bp)	18		98	

Notes: $\mu = 0.40$, $\delta = 0.08$. The multiplier $1/(1 - \Phi')$ is 2.1 in both regimes. “Bayesian” sets $\delta = 0$.

6.2 Results

Table 2 is the model’s core arithmetic. The same distrust radius that costs 18 basis points in calm costs 98 in stress: state-dependence by a factor of 5.4, produced entirely by where the marginal creditor’s density sits. Figure 1 traces the full frontier: the crisis threshold and stress spread as the fragile share runs from zero to one, for three distrust radii. The slope through any point is Proposition 3’s comparative static, and its steepening in δ is the interaction term. Figure 2 maps Proposition 4: the boundary $y^\dagger(\mu)$ at which official precision switches sign, with the destabilizing region growing in the fragile share. Figure 3 draws Proposition 5: distrust destabilizes everywhere, while equal-sized noise crosses zero and stabilizes in distress. Figure 4 prices the QT drift, whose calm-time cost of 86 minus 76 basis points understates the position: the day the year-seven shock lands, the gap is 44 basis points and permanent.

7 Empirical Designs

The model’s objects have counterparts in assembled data: the fragile share μ in the investor-base panels of Arslanalp and Tsuda (2014), distrust δ in the history of official fiscal revisions and transparency standards, disagreement in forecast dispersion, and the precision ratio α/β in the richness of a country’s private information environment. Four designs, in rising structural ambition, state the predicted signs, magnitudes, and estimating samples.

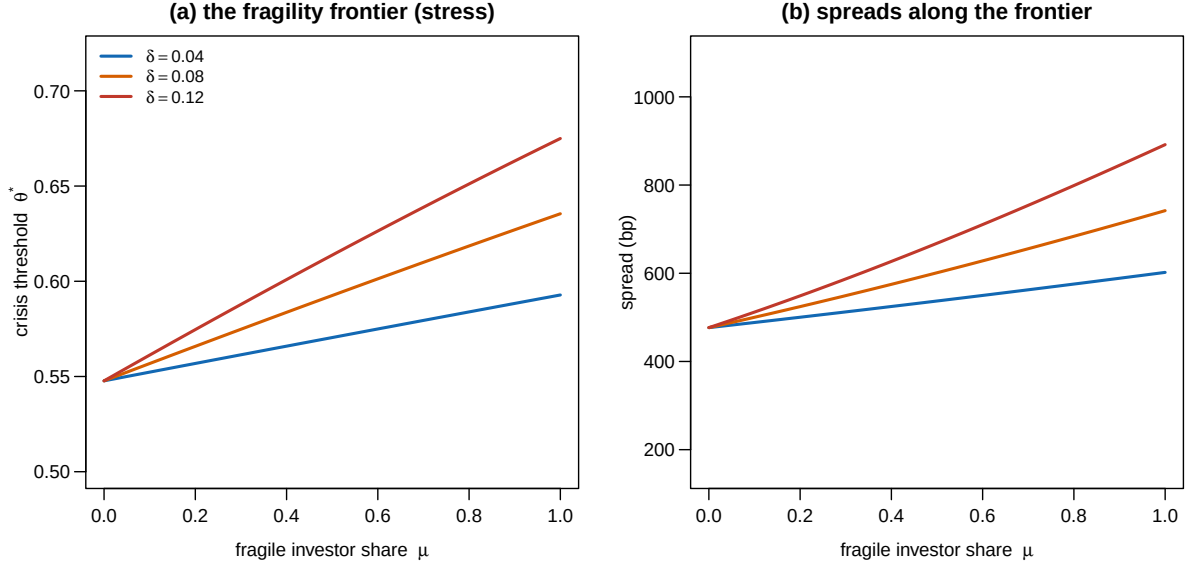


Figure 1: The fragility frontier under stress news: crisis threshold (a) and spread (b) by fragile investor share, for distrust radii $\delta \in \{0.04, 0.08, 0.12\}$.

Design 1: the interaction panel. Quarterly sovereign panel, spreads on composition and information proxies:

$$\Delta s_{i,t} = \alpha_i + \tau_t + \beta_1 D_{i,t} \times \mu_{i,t} + \beta_2 D_{i,t} \times \mu_{i,t} \times \text{Stress}_{i,t} + \beta_3 G_{i,t} \times \mu_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t}, \quad (4)$$

where D is a distrust proxy (cumulative absolute fiscal revisions in real-time data, data-standard subscription dates, statistical-integrity events), G a disagreement proxy (Consensus forecast dispersion), and μ the fragile share from investor-base data. Stress is constructed from predetermined debt service, reserve cover, maturity walls, and primary-balance forecasts, keeping the left-hand side reserved for spreads. The model predicts $\beta_1 > 0$, $\beta_2 > 0$ (Proposition 3: the premium is an interaction and it is state dependent), and, in distressed subsamples, opposite signs on distrust and disagreement (Proposition 5). The magnitudes at the calibration, 18 basis points in calm versus 98 in stress for a δ of Greek-revision size at $\mu = 0.40$, set the scale the panel recovers, with the multiplier supplied by Design 4.

Design 2: credibility events against precision events. Proposition 4 says communication policy has two instruments with different signs, which event studies can separate because the events exist separately. Credibility events are subscription to data-dissemination standards, adoption of independent fiscal councils, and external reviews. Prior evidence associates the first with lower borrowing costs (Cady, 2004), and the model predicts the effect is largest for high- μ , high- D sovereigns and strictly stabilizing. Precision events are scheduled publication of program reviews and Article IV assessments, classified by content into favorable and unfavorable news. The model predicts the *same event type* lowers spreads for sovereigns comfortably above the frontier $y^\dagger(\mu)$ of Figure 2 and raises them for sovereigns below it, with the frontier shifting

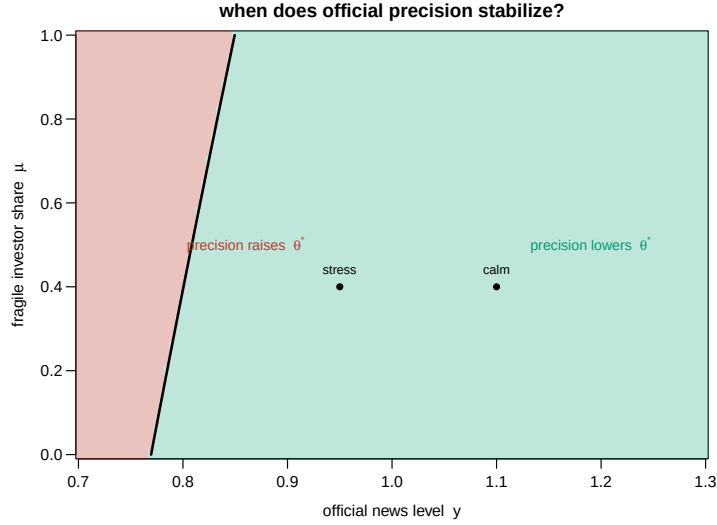


Figure 2: The sign of $d\theta^*/d\alpha$: official precision stabilizes in the green region and destabilizes in the red, separated by the boundary $y^\dagger(\mu)$ of Proposition 4. Dots mark the calm and stress calibrations at $\mu = 0.40$.

against sovereigns whose debt sits in fragile hands. The sign switch, within event type, across the fundamentals distribution, is the design’s fingerprint. Pooled event studies that average across the frontier should find the small unstable coefficients that the literature reports.

Design 3: rollover pressure in auction microdata. Crisis thresholds are about quantities, and primary auctions observe them: bid-to-cover ratios, stop-out tails, and failed or rescued auctions are the run margin in operation. The design projects auction outcomes on $D_{i,t} \times \mu_{i,t}$ with issue and window controls. The model predicts distrust widens tails and thins cover most where composition is fragile, and, by the pass-through result of Appendix C, that the sensitivity of auction outcomes to public news scales with the coordination multiplier, rising as conditions tighten.

Design 4: the distrust index, by structural inversion. Given the calibrated payoff block, measured $\mu_{i,t}$, and information parameters disciplined by dispersion data, equation (3) maps a country’s spread into an implied distrust radius $\hat{\delta}_{i,t}$: the model run backwards as a measurement device. Validation uses historical credibility events: $\hat{\delta}$ should spike at documented revisions (Greece 2009–10 above all), decay with track record, and correlate with transparency indices kept outside the inversion. With $\hat{\delta}$ in hand, the counterfactuals of Section 5, QT under distrust, CAC value by credibility environment, become country-specific calculations, and the paper’s bottom line becomes an estimated object: how much of the cross-section of sovereign spreads is trust.

Table 3 collects the tests. P4 separates the two halves of “transparency” that existing empirical work pools. P5 separates ambiguity from risk-based stories in which uncertainty is a variance. Design 1 measures the core interaction between distrust and fragile holdings, while Design 4 converts that interaction into a country-level distrust index.

Two observations already validate the mechanism and enter outside calibration. The Greek

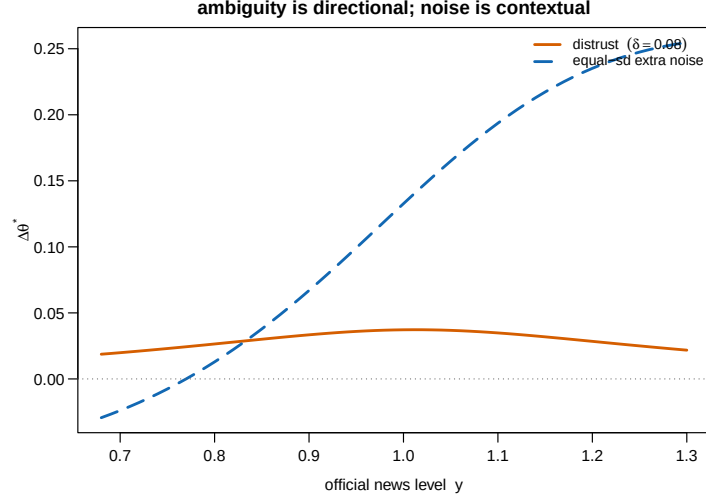


Figure 3: Proposition 5 drawn: the effect on θ^* of distrust $\delta = 0.08$ (solid) and of an equal increase in the official signal’s standard deviation (dashed), across news levels.

Table 3: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
P1: distrust $\times$ fragile share prices spreads, more in stress	18 bp $\rightarrow$ 98 bp	additive effects	spread panel
P2: distrust pricing scales with official-information dependence	loading α/β	uniform	dispersion moments
P3: QT-style composition shifts raise bad-news sensitivity	+10 calm, +44 shock (bp)	symmetric	investor-base shifts
P4: credibility events stabilize unconditionally, precision events switch sign at $y^\dagger(\mu)$	Figure 2	uniform sign	event studies
P5: disagreement and distrust have opposite signs in distress	Figure 3	same sign	stress sub-samples
P6: official announcements widen investor disagreement where credibility is low	Remark 1	narrow	survey dispersion

sequence itself is one: spreads tripled in the two quarters after the revision announcements while measured fundamentals deteriorated smoothly, which is the signature of a credibility shock repricing the threshold rather than news repricing fundamentals, and the calibration leaves that path untouched. The association of data-standard subscription with lower borrowing costs (Cady, 2004) is the other: it is the cross-sectional shadow of Proposition 4(ii), estimated two decades before this model, on data the calibration never touches.

Table 4 records the data implementation. Designs 1, 2, and 4 use public or licensed panels. Design 3 uses auction microdata from national debt-management offices, uneven in coverage and format, and assembling them is the program’s main data investment.

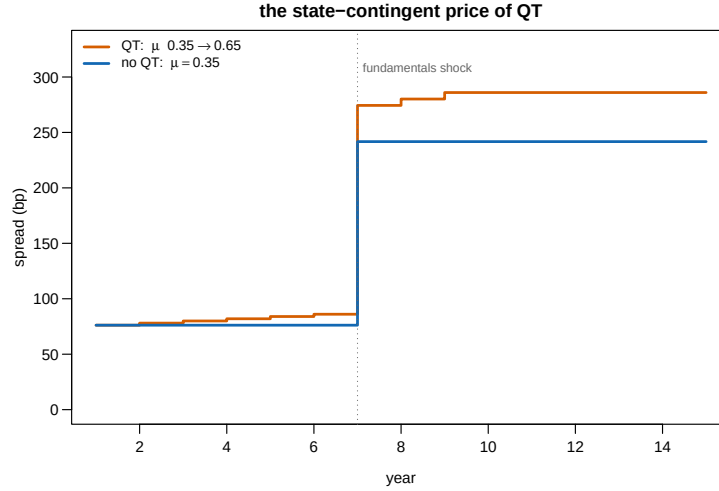


Figure 4: The state-contingent price of quantitative tightening: spread paths as the fragile share drifts from 0.35 to 0.65, against a fixed-share counterfactual. A moderate fundamentals shock arrives in year seven.

8 Conclusion

I have treated trust in official information as an economic primitive: bounded, heterogeneous across holders of the same bond, and priced through a coordination game whose threshold it moves. The model's discipline is that everything dangerous in it is measurable. Distrust has a natural scale: the magnitude of official revisions a creditor must reckon possible. Its price has a natural loading in the dependence of posterior beliefs on official sources, and its bearers have a natural census in the investor-base decompositions that already exist. Its consequences have a natural laboratory in the interaction of these objects in spreads and auction outcomes. The three theoretical results I would defend anywhere are the as-if theorem with its α/β loading, the ambiguity-proofness of global-game uniqueness, which converts fragility from a selection story into an estimable comparative static, and the decomposition of communication into precision, which helps conditionally, and credibility, which helps always.

The policy reading is a hierarchy. Buy credibility first: statistical standards, independent review, track record, and backstops with enforceable terms. These stabilize at every news level and every composition. Deploy precision only bundled with strength, because the megaphone amplifies the worst-case reading exactly in the distressed region where communication is most tempting. Manage composition as an option position: the cost of handing maturing debt to distrustful holders is invisible in calm markets and arrives, multiplied, with bad news. Treat contractual design, CACs and their statutory relatives, as trust infrastructure: compressing worst-case losses builds credibility into the contract itself.

The next extension is learning: letting creditors update δ from the record of official statements turns credibility into a slow-moving state with hysteresis of its own, a reputational asset built in decades and spent in one October. The Greek episode suggests the asymmetry is severe. The

Table 4: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Implementation
Investor base (Arslanalp and Tsuda, 2014)	AE and EM sovereigns, 2004–	quarterly	public	observed	in hand
Real-time fiscal out-turns	IMF vintages, WEO archive	semiannual	public	observed	WEO archive build in hand
Data-standard subscriptions	IMF dissemination standards	event	public	observed	in hand
Forecast dispersion	Consensus-type surveys	monthly	licensed	observed	licensed panel
Bond and CDS spreads	liquid sovereigns	daily	licensed	observed	licensed panel
Auction microdata	debt-office releases	per auction	uneven	observed where released	debt-office sample
IMF review dates	Article IV, program archive	event	public	observed	in hand

model prices the level of trust. The dynamics of losing and rebuilding it are the next margin.

A Proofs

A.1 Proof of Lemma 1

In threshold strategies with cutoffs (x_k^*) , the run mass $\sum_k \mu_k \Phi(\sqrt{\beta}(x_k^* - \theta))$ is continuous and strictly decreasing in θ while capacity θ is strictly increasing, so there is a unique crossing θ^* and the crisis event is exactly $\{\theta \leq \theta^*\}$: a fundamental event, fixed given the profile. A creditor of type k contemplating rolling evaluates $(1 - P)R + P\mathcal{R}$ with $P = \mathbb{P}_b(\theta \leq \theta^* \mid x_i, y)$ under bias model b . The posterior mean $[\alpha(y - b) + \beta x_i]/(\alpha + \beta)$ is decreasing in b , so P is increasing in b and the max-min creditor evaluates at $b = \delta_k$. His entire decision problem is then that of a Bayesian with public signal $y - \delta_k$. The run payoff $1 - \kappa$ is constant across models, so a single minimization suffices. ■

A.2 Proof of Proposition 1

With posterior $\theta \mid x \sim N(m_k(x), (\alpha + \beta)^{-1})$, indifference $\mathbb{P}(\theta \leq \theta^* \mid x_k^*) = \bar{p}$ requires $m_k(x_k^*) = \theta^* - \Phi^{-1}(\bar{p})/\sqrt{\alpha + \beta}$, a quantity independent of k . Since $m_k(x) = [\alpha(y - \delta_k) + \beta x]/(\alpha + \beta)$, solving for x_k^* gives $x_k^* = x_B^* + (\alpha/\beta)\delta_k$ where x_B^* solves the $\delta = 0$ case. Multiplying the cutoff gap by $\sqrt{\beta}$ gives the run-propensity shift $(\alpha/\sqrt{\beta})\delta_k$ in (3). ■

A.3 Proof of Proposition 2

Substituting cutoffs into the crossing condition gives (3), whose right side has derivative $(\alpha/\sqrt{\beta})\sum_k \mu_k \phi(g + \frac{\alpha}{\sqrt{\beta}}\delta_k) \leq (\alpha/\sqrt{\beta})\phi(0) = (\alpha/\sqrt{\beta})/\sqrt{2\pi} < 1$ under the stated condition, uniformly in (μ, δ) : the map is a contraction on $[0, 1]$ after noting its range lies in $(0, 1)$, so the fixed point exists and is unique, and the associated cutoff profile is the unique monotone equilibrium. The extension from monotone strategies to full rationalizability follows the standard global-games iterated-dominance argument (Morris and Shin, 2003), which the parallel-shift structure of Lemma 1 leaves intact. ■

A.4 Proof of Proposition 3

Let $s = \alpha/\sqrt{\beta}$, let $A(\theta, \delta) = \Phi(g(\theta) + s\delta)$, and let $B(\theta) = \Phi(g(\theta))$, where $g(\theta)$ is the affine index in (3). With two types, the fixed point is

$$H(\theta, \mu, \delta) = (1 - \mu)B(\theta) + \mu A(\theta, \delta) - \theta = 0.$$

Write $\Phi'_\theta = s[(1 - \mu)\phi_B + \mu\phi_F]$. The uniqueness condition gives $H_\theta = \Phi'_\theta - 1 < 0$. Implicit differentiation in μ gives

$$\frac{d\theta^*}{d\mu} = -\frac{H_\mu}{H_\theta} = \frac{A(\theta^*, \delta) - B(\theta^*)}{1 - \Phi'_\theta} > 0,$$

because $\delta > 0$ shifts the fragile cutoff to the right. Differentiating the numerator in δ gives $s\phi_F > 0$. Differentiating the denominator gives $-s^2\mu\phi'_F$, which is positive exactly when $g + s\delta < 0$,

so both the numerator and the denominator movement raise $d\theta^*/d\mu$ in that region. Finally,

$$\frac{d\theta^*}{d\delta} = \frac{\mu s \phi_F}{1 - \Phi'_\theta} > 0.$$

Its pass-through to the ex ante crisis probability multiplies by $\phi((\theta^* - y)\sqrt{\alpha})\sqrt{\alpha}$, which is largest when fundamentals news places mass near the threshold. ■

A.5 Proof of Proposition 4

Differentiate (3) in α : with g as defined, $\partial g/\partial\alpha = [(\theta^* - y) - \Phi^{-1}(\bar{p})/(2\sqrt{\alpha + \beta})]/\sqrt{\beta}$ and $\partial(\frac{\alpha}{\sqrt{\beta}}\delta_k)/\partial\alpha = \delta_k/\sqrt{\beta}$, so

$$\frac{d\theta^*}{d\alpha} = \frac{\sum_k \mu_k \phi_k [(\theta^* - y) - \frac{\Phi^{-1}(\bar{p})}{2\sqrt{\alpha + \beta}} + \delta_k]}{\sqrt{\beta}(1 - \Phi')},$$

whose sign is that of $y^\dagger - y$ with $y^\dagger$ as stated. For $\bar{p} < 1/2$, $\Phi^{-1}(\bar{p}) < 0$, so the middle term is positive and $y^\dagger > \theta^*$ even at $\delta = 0$. The distrust term adds the ϕ -weighted mean of δ_k . Part (ii) is the δ_k -derivative computed in Proposition 3(iii), positive under the stated derivative. ■

A.6 Proof of Proposition 5

(i) is Proposition 3(iii). (ii) As $\alpha \rightarrow 0$, $g \rightarrow -\Phi^{-1}(\bar{p})$ and the shifts vanish, so the right side of (3) converges uniformly to the constant $\Phi(-\Phi^{-1}(\bar{p})) = 1 - \bar{p}$: the fixed point converges to the anchor $1 - \bar{p}$ regardless of y , μ , δ . By continuity of $\theta^*(\alpha)$, the total effect of sufficiently degraded precision is $1 - \bar{p} - \theta^*$: negative, stabilizing, wherever distress has pushed θ^* above the anchor, and positive otherwise. (iii) In states with $\theta^* > 1 - \bar{p}$, (i) and (ii) have opposite signs, so the variance effect changes sign across states while the distrust comparative static remains monotone. ■

B Derivations: Pricing and Pass-Through

Spreads. With risk-neutral pricing at a zero reference rate, a bond promising R with crisis probability P and recovery $\mathcal{R}$ carries spread $s = P(1 - \mathcal{R})/[1 - P(1 - \mathcal{R})]$ per period, the mapping used throughout. For small P it is $P \times$ loss given default. Endogenizing R through an issuance condition adds a second fixed point that raises all fragility statics (a higher threshold raises the promised yield, which through $\bar{p}$ raises the threshold again). The reported magnitudes are conservative in omitting it.

News pass-through and the multiplier. Differentiating (3) in y gives $d\theta^*/dy = -\Phi'/(1 - \Phi')$ and hence

$$\frac{dP}{dy} = \phi((\theta^* - y)\sqrt{\alpha})\sqrt{\alpha} \left(\frac{d\theta^*}{dy} - 1 \right) = - \frac{\phi((\theta^* - y)\sqrt{\alpha})\sqrt{\alpha}}{1 - \Phi'} :$$

the coordination multiplier scales the pass-through of public news to crisis risk one-for-one. Every announcement-window regression in Section 7 estimates a coefficient carrying this factor, which is how Design 4’s structural step separates the direct density from the amplification.

C Dynamics and the Doom Loop

Composition evolves by accounting: with runoff fraction q_t of the official portfolio’s maturing stock, $\mu_{t+1} = \mu_t + q_t(1 - \mu_t)\omega_t$, where ω_t is the official share of maturing debt. The QT experiment in the text uses a linear drift matching an eight-year normalization. The debt-service feedback closes fundamentals: $y_{t+1} = \bar{y}_{t+1} - \varsigma(s_t - s^{ref})$, service costs subtracting from next period’s capacity news. Combining with the pass-through result above, the loop is locally explosive when

$$\varsigma \times \frac{\phi((\theta^* - y)\sqrt{\alpha})\sqrt{\alpha}(1 - \mathcal{R})}{1 - \Phi'} > 1,$$

a condition that tightens mechanically as stress raises the density term and as composition and distrust raise Φ' : the same statics that price spreads shrink the distance to the spiral. This is the Calvo (1988) instability with the coordination multiplier made explicit. Bocola and Dovis (2019) quantify the Bayesian analogue. The text’s experiments hold $\varsigma = 0$ to isolate the state-contingent spread gap that any positive ς would amplify.

D Parameter Discipline and Sensitivity

Table 5 records where each parameter enters, its units, its source, the range examined, and what moves. At $\delta = 0$ the model is the standard Bayesian global game and the distinctive predictions P1, P4’s distrust term, P5, and P6 return to zero. As $\alpha/\sqrt{\beta}$ approaches 2.51, the multiplier grows and the distance to selection becomes the relevant statistic, with the margin discussed in Section 2.

E Data Appendix

Investor base. Arslanalp and Tsuda (2014) and successor vintages: quarterly holdings of sovereign debt by foreign official, foreign bank, foreign nonbank, domestic bank, domestic nonbank, and central bank, for advanced and emerging sovereigns. The fragile share μ aggregates the price-sensitive private categories, with maturity weighting where debt-office data permit.

Distrust proxies. Real-time versus final fiscal outturns (cumulative absolute revision within a window), data-standard subscription dates (Cady, 2004), and documented statistical-integrity events, with the Greek revisions of 2009–10 as the anchor case (European Commission, 2010).

Disagreement proxies. Consensus-type forecast dispersion for deficits and growth, bid-ask spreads, and, where available, CDS-implied volatility smiles.

Table 5: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range exam- ined	Effect of the range
R, κ	eq. (1)	face value	gross yield, fire-sale discount	joint shift	shift $\bar{p}$ jointly
$\mathcal{R} = 0.60$	eq. (1)	recovery rate	Cruces and Trebesch (2013)	0.60 to 0.70 (CAC exp.)	stress spread 575 to 128 bp
$\bar{p} = 27.3\%$	eq. (3)	probability	implied	follows $\mathcal{R}$	anchor $1 - \bar{p}$ moves
σ_x, σ_y	eq. (3)	fundamentals units	calibrated input, Design 4	12-cell grid, σ_x 0.10 to 0.30, σ_y 0.25 to 0.50	premia 0 to 51 / 0 to 98 bp, 3 cells cross uniqueness
$\alpha/\sqrt{\beta} = 1.35$	uniqueness	unitless	implied	up to bound 2.51	multiplier grows, then multiplicity
$\delta = 0.08$	eq. (2)	fundamentals units	Greek-revision scale, Design 4	0.04 to 0.12 (Fig. 1)	premium 18 to 98 bp across states
$\mu = 0.40$	eq. (3)	maturing-debt share	Arslanalp and Tsuda (2014)	0 to 1 (Fig. 1), 0.35 to 0.65 (QT)	QT premium 10 to 44 bp
y calm/stress	eq. (3)	fundamentals units	internal, spread 79 bp	calm full axis in figures	state dependence of every result

Notes: the payoff block is estimated, the information block is disciplined by dispersion data in Design 4, and the structural inversion delivers the country-level distrust radius.

Market outcomes. Sovereign bond and CDS spreads at daily frequency for announcement windows and quarterly for the panel. Primary-auction results (bid-to-cover, stop-out tails, cancellations) from debt-management offices, and IMF Article IV and program-review publication dates from the IMF archive.

F Extensions

Heterogeneous payoffs. Giving types distinct $(R_k, \mathcal{R}_k, \kappa_k)$, balance-sheet borrowing and mark-to-market sensitivity as payoff heterogeneity, yields type-specific hurdles $\bar{p}_k$ entering through $\Phi^{-1}(\bar{p}_k)$ in each cutoff. Every result survives with type-specific constants, and the fragile type of the text is then jointly distrustful and low-hurdle, which is how the data likely mix the two. Separating preference-based from information-based fragility is possible in principle through P6, since only the information channel produces announcement-fanning.

Smooth ambiguity. Replacing max-min with smooth ambiguity aversion turns the worst-case shift δ_k into an endogenous pessimism weight varying with stakes. The as-if structure survives to first order, with δ_k reinterpreted as the local ambiguity premium. The max-min case is the tractable bound.

Learning about bias. Letting creditors update the bias set from realized revisions makes δ_t a state with its own persistence: credibility as reputational capital. The asymmetry between

slow accumulation and one-announcement destruction suggests trap dynamics complementary to those studied here.

Dynamic runs. Staggered maturities in the sense of He and Xiong (2012) let today's rollover outcome inform tomorrow's, adding an intertemporal coordination motive to the static one. The composition state μ_t then also indexes the speed at which a run can build. The static threshold is the natural building block for that model.

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